RAK12501 WisBlock GNSS Location Module Datasheet

Overview

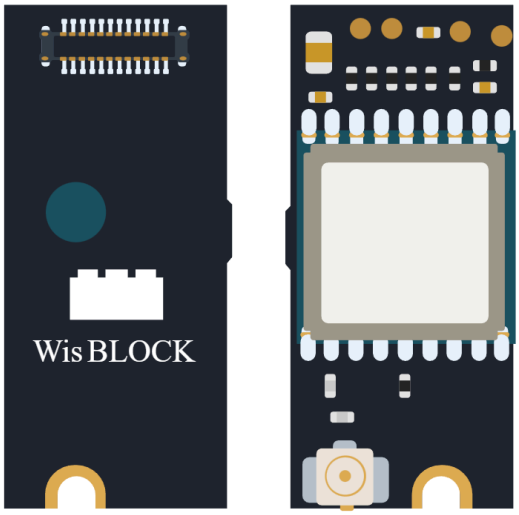


Figure 1: RAK12501 WisBlock GNSS Location Module

Description

The RAK12501 WisBlock GNSS Location Module is designed to be part of a production-ready IoT solution in a modular way and must be combined with a WisBlock Core and a Base module.

Equipped with the Quectel L76K, the RAK12501 supports a wide variety of satellite data protocols such as GPS, GLONASS, QZSS, and BeiDou. This ensures the retrieval of precise location data. The module features exceptional performance, high sensitivity, and minimal acquisition time. A very suitable module for your low-power IoT solution needs.

Features

- **Module Specification**
  - Uses the very accurate GNSS Module: **Quectel L76K**
  - Location Accuracy of ±2.5 meter
  - Velocity Accuracy of ±0.1 m/s
  - GPS, GLONASS, QZSS, and BeiDou Satellite support
  - Serial communication to WisBlock Core support
  - 1 Hz Update Rate
  - Location Fix:
    - Cold Start: 35 seconds
    - Hot Start: 1 second
  - Operating Voltage: 3.3 V
  - Operating Current: < 25 mA
- **Size**
  - Module Size: 10 mm x 23 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK12501 module on a WisBlock Base board. The RAK12501 module can be mounted on the slots: **SLOT A or SLOT D**.

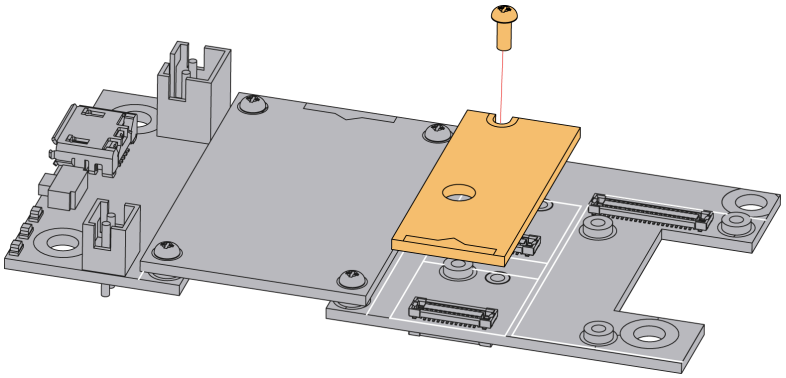


Figure 2: RAK12501 Module mounting mechanism

Hardware

The hardware specification is categorized into five parts that cover the chipset and pinouts and the corresponding functions and diagrams of the board. It also presents the parameters and their standard values in terms of electrical and mechanical.

Chipset

The RAK12501 utilizes a very accurate Quectel L76K chip. See the manufacturer's Quectel L76K Page for more details.

Vendor	Part Number
Quectel	L76K

Pin Definition

The RAK12501 WisBlock GNSS Location Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK12501 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

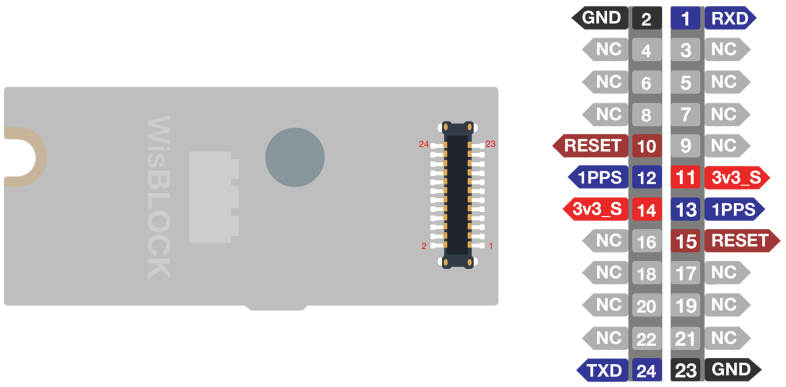


Figure 3: RAK12501 GNSS Module pinout

Pin Number	Description	Pin Number	Description
1	RXD	2	GND
3	NC	4	NC
5	NC	6	NC

Pin Number	Description	Pin Number	Description
7	NC	8	NC
9	NC	10	RESET
11	3V3_S	12	1PPS
13	1PPS	14	3V3_S
15	RESET	16	NC
17	NC	18	NC
19	NC	20	NC
21	NC	22	NC
23	GND	24	TXD

NOTE

Only the UART related pins, VDD, and GND are connected to this module. WB_IO2 is used to control the power supply of the module.

Electrical Characteristics

Recommended Operating Conditions

The table below shows the recommended operating conditions for the RAK12501 WisBlock GNSS Location Module:

Symbol	Description	Min.	Nom.	Max.	Unit
V_{DD}	Power supply for the module	2.8	3.3	4.3	V
$I_{Acquisition}$	Power Acquisition	-	25	-	mA
$I_{Tracking:}$	Power Tracking	-	18	-	mA
$I_{Saving:}$	Power Saving	-	7	-	uA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and mechanical drawing of the RAK12501 module.

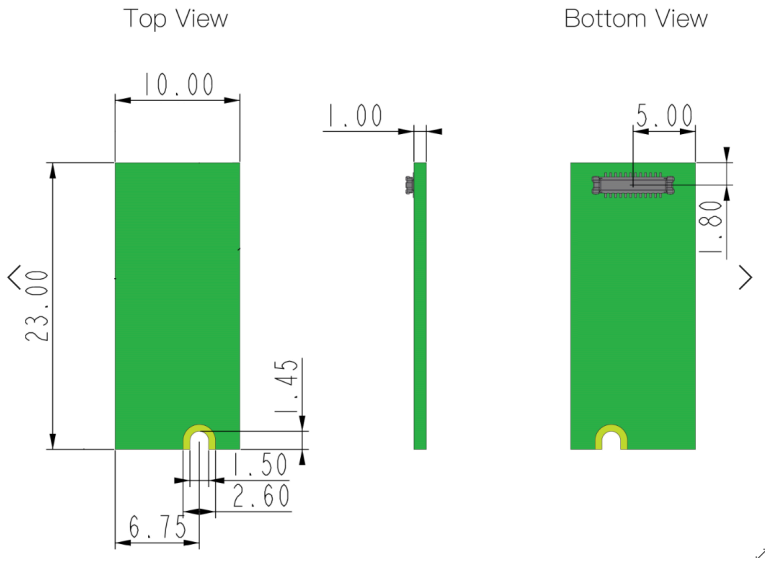


Figure 4: RAK12501 board dimensions

WisConnector PCB Layout

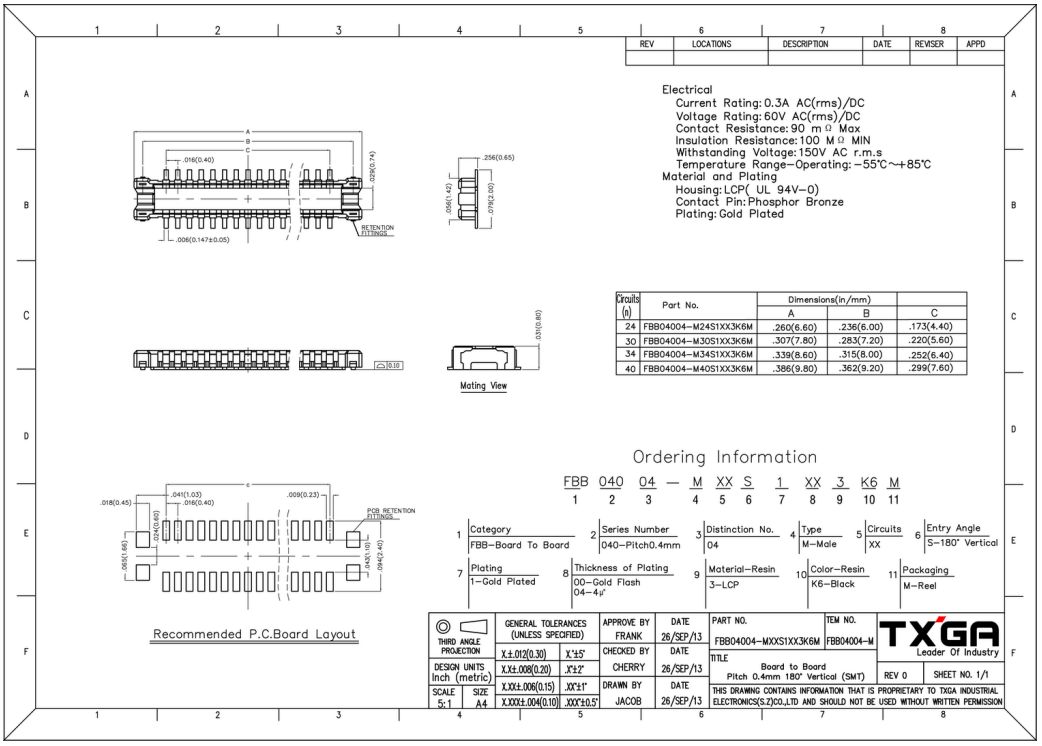


Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

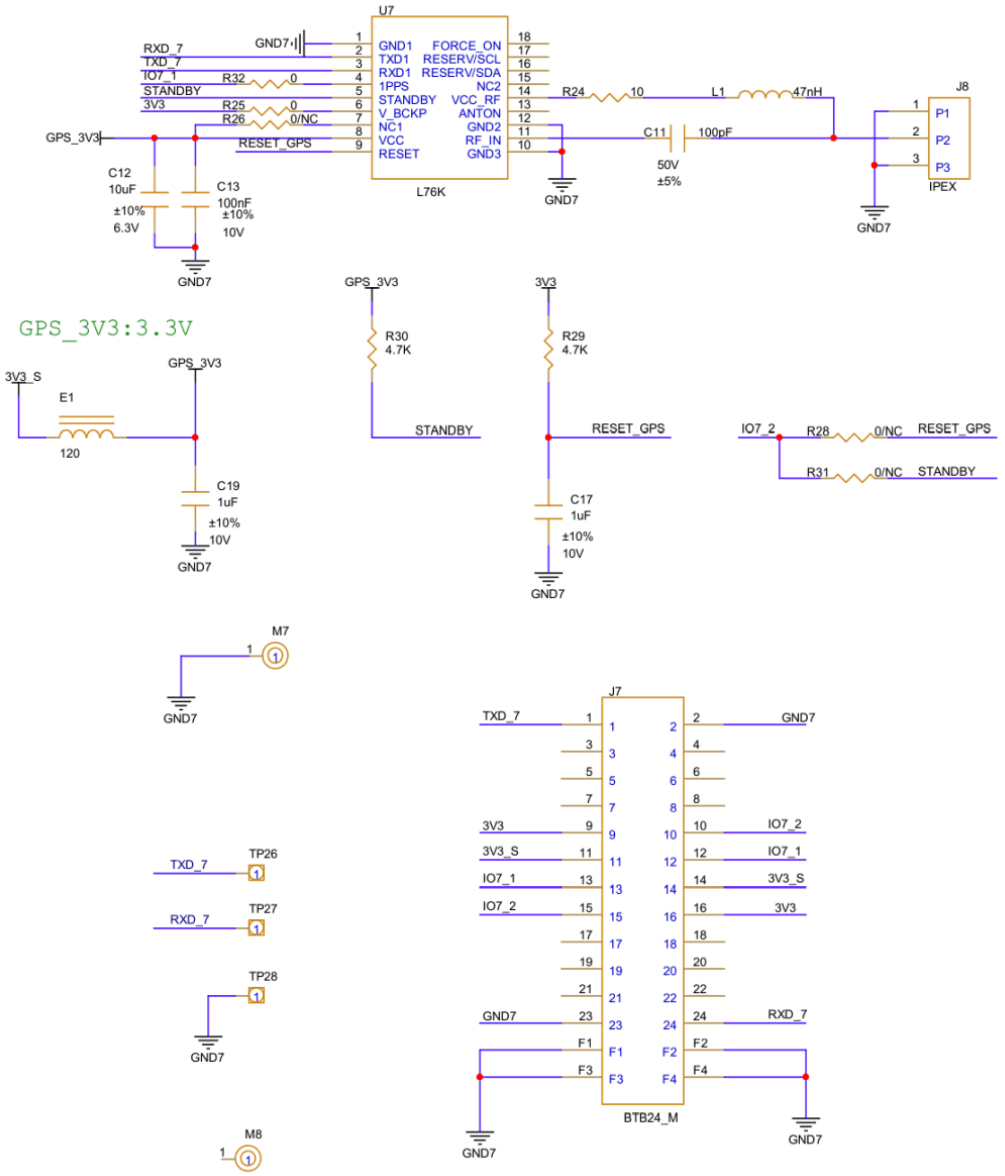


Figure 6: RAK12501 schematic diagram